

Gauge Symmetry Structure of the WIN Loop Operator and the DNLS Hartree Hamiltonian on the 14 Mode Multiplet

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We establish the gauge symmetry structure of two operators acting on the 14-dimensional multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$: the WIN loop operator $M_{\text{loop}} = T L_{\text{EFT}} L_{\text{Core}} L_{\text{DME}} L_{\text{FDLM}}$ (the five-layer processing composition of the WIN Paradigm), and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ with $K[a, b] = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2$. Both operators are diagonal or diagonal-plus-rank-one in the mode basis, yet their gauge-symmetry content is not obvious. We compute the intersection of $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ with the commutant of each operator and find: (i) $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional abelian subalgebra of $\mathfrak{g}_{\text{SM}}$, contained in the span of the Cartan generators and the isotypic projectors and rotated by angles 26.2° , 42.2° , 90.0° from the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$; (ii) $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is a 10-dimensional Lie subalgebra isomorphic to $\mathfrak{su}(2) \oplus \mathfrak{u}(1)^7$, containing the full $\mathfrak{su}(2)_L$ factor, all five isotypic projectors, and a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$ rotated by angles 33.4° and 77.8° from the standard Cartan $\langle \lambda^3, \lambda^8 \rangle$. The $\mathfrak{su}(2)_L$ preservation is exact and structural: it holds for every value of the DNLS interaction strength U , because the two B_1 isotypic vectors pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ with opposite signs, producing exact cancellation in the sum of densities. Neither result depends on the DME constant γ , the warp factor kL , or any other calibrated input. All claims are verified by explicit computation at machine precision in IEEE-754 double precision (`numpy float64`).

I. INTRODUCTION

The WIN Paradigm [1, 2] derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Hodge complex on Λ carries a D_4 action, and at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian the content decomposes as

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (1)$$

of total dimension 14 [3, 4]. The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ embeds in the D_4 commutant on this space, with $\mathfrak{su}(3)_c$ acting on the A_1 multiplicity and $\mathfrak{su}(2)_L$ on the B_1 multiplicity [4].

Two operators act naturally on the 14-mode multiplet. The first is the WIN loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}, \quad (2)$$

a composite of five processing layers [5, 6]. The second is the DNLS Hartree Hamiltonian

$$H_{\text{DNLS}} = 4\mathbb{I} + UK, \quad K[a, b] = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (3)$$

the mean-field approximation of the discrete nonlinear Schrödinger substrate Hamiltonian [7].

A natural structural question is: which subalgebra of $\mathfrak{g}_{\text{SM}}$ is preserved by each operator? This is the Connection Theorem question of the WIN Paradigm. We address it here for both operators and find distinct answers. The loop preserves a rotated 3-dimensional abelian subalgebra. The DNLS Hamiltonian preserves a 10-dimensional subalgebra containing the full electroweak factor and a rotated abelian subalgebra of $\mathfrak{su}(3)_c$.

The paper is organized as follows. Section II sets notation and fixes the mode basis, class partition, and gauge embedding. Section III proves the loop result. Section IV proves the DNLS result, including the structural mechanism for exact $\mathfrak{su}(2)_L$ preservation. Section V interprets the results.

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II. SETUP

A. The 14-mode multiplet

The discrete 5-point Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (4)$$

The eigenvalue $\lambda = 4$ has multiplicity 14, realized on the set $\mathcal{M}$ consisting of $\{(0, 4), (4, 0)\}$ (orbit $\mathcal{O}_0$), $\{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 1), (7, 5)\}$ (orbit $\mathcal{O}_1$), and $\{(2, 2), (2, 6), (6, 2), (6, 6)\}$ (orbit $\mathcal{O}_2$).

The real mode vectors are

$$\phi_{(n_1, n_2)}(x, y) = \frac{\cos\left(\frac{2\pi(n_1 x + n_2 y)}{8}\right)}{\|\cos(\cdots)\|_2}, \quad (5)$$

normalized so that $\sum_{x, y} |\phi_m(x, y)|^2 = 1$. The inverse participation ratio $\text{IPR}_m = \sum_{x, y} |\phi_m(x, y)|^4$ takes the value $2/128, 3/128, 4/128$ on $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ respectively [8].

B. Class partition and loop eigenvalues

The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$, built from the bilinear $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$ [5], has spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$. The 14 modes at $\lambda = 4$ partition into three classes by frequency:

$$A = \{(0, 4), (4, 0), (6, 6)\}, \quad \omega_A = 2\sqrt{2}, \quad (6)$$

$$B = \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\}, \quad \omega_B = 6\sqrt{2}, \quad (7)$$

$$C = \{(5, 7), (2, 2), (2, 6), (6, 2)\}, \quad \omega_C = 0. \quad (8)$$

The loop operator M_{loop} is diagonal in the mode basis with three distinct eigenvalues $\lambda_A, \lambda_B, \lambda_C$ on the three classes [5, 6].

Remark 1. The class partition does not respect the D_4 orbit partition: it splits $\mathcal{O}_2$ across A and C and separates one mode from $\mathcal{O}_1$ into class C . This misalignment is the structural origin of the loop's symmetry-breaking pattern.

C. Gauge embedding

Let $\{u_0, u_1, u_2\}$ be an orthonormal basis of the A_1 isotypic $V_{A_1} = \mathbb{C}^3$, chosen as the uniform real superpositions over $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ respectively. Let $\{v_0, w_1\}$ be an orthonormal basis of the B_1 isotypic $V_{B_1} = \mathbb{C}^2$, chosen as follows:

$$v_0 = \frac{1}{\sqrt{2}}(\phi_{(0,4)} - \phi_{(4,0)}), \quad (9)$$

$$w_1 = \frac{1}{\sqrt{8}}(\phi_{(1,3)} + \phi_{(1,5)} - \phi_{(3,1)} - \phi_{(3,7)} - \phi_{(5,7)} + \phi_{(7,3)} + \phi_{(7,5)} - \phi_{(5,1)}). \quad (10)$$

Both vectors are real, orthogonal, and transform in the B_1 representation of D_4 .

The $\mathfrak{su}(3)_c$ generators are $T_c^a = U_{A_1} \lambda^a U_{A_1}^T$, $a = 1, \dots, 8$, and the $\mathfrak{su}(2)_L$ generators are $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$, $i = 1, 2, 3$. The five $\mathfrak{u}(1)$ generators are the isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$. We write $g_1, \dots, g_{16}$ for the sixteen generators in the order $(\lambda^1, \dots, \lambda^8, \sigma^1, \sigma^2, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E)$.

D. Numerical environment

All computations reported in this paper were performed with NumPy arrays in IEEE-754 double precision (`float64`), i.e. machine epsilon $\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$ and 53-bit mantissa. Singular-value spectra of the commutator matrices were computed via the LAPACK `gesdd` driver. A numerical rank threshold of 10^{-10} was used throughout, but the results are insensitive to the threshold: the reported singular-value gaps are 13 to 15 orders of magnitude, so any threshold in $[10^{-6}, 10^{-14}]$ produces the same rank and the same nullspace dimension. Reproducibility of all figures at this precision has been verified.

III. THE LOOP OPERATOR

Theorem 2 (Loop gauge-symmetry structure). *Let M_{loop} be the WIN loop operator (2) acting on $\Omega^0|_{\lambda=4}$. The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a three-dimensional abelian subalgebra of $\mathfrak{g}_{\text{SM}}$. Its generators are supported on the span of $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$ and contain no root generator of $\mathfrak{su}(3)_c$ or $\mathfrak{su}(2)_L$. The principal angles between this subalgebra and the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ are 26.2° , 42.2° , and 90.0° .*

Proof. Let P_A, P_B, P_C be the class projectors. Since $\lambda_A, \lambda_B, \lambda_C$ are pairwise distinct, $M_{\text{loop}} = \lambda_A P_A + \lambda_B P_B + \lambda_C P_C$, and an operator commutes with M_{loop} iff it commutes with each of P_A, P_B, P_C individually.

Form the 196×16 matrix $\mathcal{C}$ whose i -th column is $\text{vec}([M_{\text{loop}}, g_i])$. Its singular value decomposition has rank 13, with singular values

$$\{9.86, 9.86, 9.86, 9.86, 7.05, 7.05, 6.13, 5.40, 5.40, 4.74, 1.42, 1.35, 1.02\} \times 10^{-1}, \quad (11)$$

followed by three singular values of order 10^{-16} . The gap between the 13th and 14th singular values is 15 orders of magnitude, so $\dim \ker(\mathcal{C}) = 16 - 13 = 3$.

The three nullspace vectors, computed by SVD with any threshold in the range $[10^{-6}, 10^{-14}]$, are real linear combinations of $\lambda^3, \lambda^8, \sigma^3$ and the five projectors. No generator with a root component ($\lambda^{1,2,4,5,6,7}$ or $\sigma^{1,2}$) appears in any nullspace vector. The brackets $[u_i, u_j]$ for the three nullspace generators u_1, u_2, u_3 have norm 2.3×10^{-16} , confirming the algebra is abelian.

Principal angles to the standard Cartan follow from the SVD of the 3×3 matrix $Q_u^T Q_{\text{Cartan}}$, giving the stated values. The 90.0° angle reflects the fact that the loop intersection contains a component orthogonal to the Cartan. $\square$

Corollary 3. *The loop preserves Cartan weights but not the standard Cartan itself. It preserves neither hypercharge, nor weak isospin, nor color charge.*

Remark 4 (On the principal angles). The angles 26.2° , 42.2° , and 90.0° are numerically computed as arccos of the singular values of the 3×3 overlap matrix $Q_u^T Q_{\text{Cartan}}$. Because the intersection is defined by the kernel of an integer-valued commutator matrix (the class projectors and the Cartan generators of $\mathfrak{su}(3)_c$, $\mathfrak{su}(2)_L$ have rational entries in the mode basis), the nullspace basis vectors are themselves algebraic numbers. The principal angles are therefore expected to satisfy polynomial equations of low degree, and a closed-form trigonometric representation may exist. We defer a root-of-unity analysis to future work.

IV. THE DNLS HARTREE HAMILTONIAN

Theorem 5 (DNLS gauge-symmetry structure). *Let $H_{\text{DNLS}} = 4\mathbb{I} + UK$ be the DNLS Hartree Hamiltonian (3) on the 14-mode multiplet, with U arbitrary real. The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is a 10-dimensional Lie subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2) \oplus \mathfrak{u}(1)^7$. Its decomposition into $\mathfrak{g}_{\text{SM}}$ -components is*

$$\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}) = \mathfrak{su}(2)_L \oplus \langle P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E \rangle \oplus \mathfrak{c}, \quad (12)$$

where $\mathfrak{c}$ is a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$ with principal angles 33.4° and 77.8° to the standard Cartan $\langle \lambda^3, \lambda^8 \rangle$. The color generators $\lambda^{1,\dots,7}$ individually do not commute with H_{DNLS} .

The proof proceeds in three steps: the reduction to the class partition, the direct computation of $\mathfrak{su}(3)_c \cap \text{Comm}(H_{\text{DNLS}})$, and the structural argument for exact $\mathfrak{su}(2)_L$ preservation.

A. Reduction to the class partition

The commutator $[H_{\text{DNLS}}, g_i]$ is a 14×14 matrix. For the 16 gauge generators, the singular-value spectrum of the stacked 392×16 matrix has rank 6, followed by 10 singular values of order 10^{-15} . The gap is 13 orders of magnitude, so the intersection is 10-dimensional. Explicitly, the singular values are

$$\{1.93, 1.93, 1.57, 1.57, 3.60, 3.60\} \times 10^{-1}, \quad (13)$$

$$\{1.18, 0.78, 0.78, 0.58, 0.45, 0.32\} \times 10^{-15}. \quad (14)$$

B. The color subalgebra

Projecting the 10-dimensional nullspace onto the $\mathfrak{su}(3)_c$ generators (rows 1–8 of the 16-generator basis), the SVD of the 8×10 projected matrix has rank 2. The two surviving basis vectors s_0, s_1 are real combinations of $\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$ with coefficients

$$s_0 = -0.555\lambda^1 + 0.656\lambda^3 - 0.249\lambda^4 - 0.213\lambda^6 + 0.393\lambda^8, \quad (15)$$

$$s_1 = 0.158\lambda^3 + 0.223\lambda^4 + 0.891\lambda^6 + 0.363\lambda^8. \quad (16)$$

The bracket norm $\|[s_0, s_1]\| = 2.3 \times 10^{-14}$, confirming the 2-dimensional $\mathfrak{su}(3)_c$ subalgebra is abelian. Its principal angles to the standard Cartan $\langle \lambda^3, \lambda^8 \rangle$ are 33.4° and 77.8° .

C. Structural mechanism for exact $\mathfrak{su}(2)_L$ preservation

The $\mathfrak{su}(2)_L$ generators $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ are supported entirely on V_{B_1} . We show that the compression $K_{B_1} = U_{B_1}^T K U_{B_1}$ vanishes identically.

Lemma 6 (Doubled-momentum pairing). *Let ϕ_m be a real mode vector on Λ with momentum labels $(n_1, n_2) \in \mathbb{Z}_8^2$. Then*

$$|\phi_m(x, y)|^2 = \frac{1}{2} \left(1 + \cos \left(\frac{\pi(2n_1x + 2n_2y)}{4} \right) \right) / \|\phi_m\|_2^2. \quad (17)$$

Two modes m, m' with $(n_1, n_2) \neq (n'_1, n'_2)$ have overlapping density only if the doubled momenta $(2n_1, 2n_2)$ and $(2n'_1, 2n'_2)$ are congruent modulo 8.

Proof. Direct computation using $\cos^2 \theta = (1 + \cos 2\theta)/2$. $\square$

Lemma 7 (Cancellation on V_{B_1}). *The two B_1 basis vectors pair modes with the same doubled momentum with opposite signs:*

- v_0 : $(0, 4) \leftrightarrow (4, 0)$, both doubling to $(0, 0)$, with coefficients $+1/\sqrt{2}$ and $-1/\sqrt{2}$.
- w_1 : eight $\mathcal{O}_1$ modes pair as $(1, 3) \leftrightarrow (5, 7)$, $(1, 5) \leftrightarrow (5, 1)$, $(3, 1) \leftrightarrow (7, 5)$, $(3, 7) \leftrightarrow (7, 3)$, each pair with same doubled momentum and opposite signs $\pm 1/\sqrt{8}$.

Consequently $\sum_{m \in \mathcal{O}_1} c_m \phi_m(x, y)^2 = 0$ for all (x, y) , where c_m are the w_1 coefficients. The same holds for v_0 on $\mathcal{O}_0$.

Proof. Direct computation. The $\mathcal{O}_1$ pairing is explicit in the list of coefficients; the $\mathcal{O}_0$ pairing follows from the antisymmetry under the reflection $(n_1, n_2) \rightarrow (n_2, n_1)$. $\square$

Proof of exact $\mathfrak{su}(2)_L$ preservation. By Lemma 7,

$$K_{B_1}[0, 0] = \sum_x \left(\sum_{m \in \mathcal{O}_1} c_m \phi_m(x)^2 \right)^2 = \sum_x 0 = 0, \quad (18)$$

and likewise $K_{B_1}[1, 1] = 0$. The off-diagonal

$$K_{B_1}[0, 1] = \sum_x \left(\sum_{m \in \mathcal{O}_1} c_m \phi_m^2 \right) \left(\sum_{m' \in \mathcal{O}_0} c_{m'} \phi_{m'}^2 \right) = 0 \quad (19)$$

for the same reason. Hence $K_{B_1} = 0$ as a 2×2 matrix.

Since the $\mathfrak{su}(2)_L$ generators are $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$, we have $KT_L^i = KU_{B_1} \sigma^i U_{B_1}^T = U_{B_1} K_{B_1} \sigma^i U_{B_1}^T = 0$. Similarly $T_L^i K = 0$. Therefore $[K, T_L^i] = 0$, and by linearity $[H_{\text{DNLS}}, T_L^i] = 0$ for all $U \in \mathbb{R}$. $\square$

D. Why $\mathfrak{su}(3)_c$ is broken

The A_1 basis vectors are same-sign superpositions: $u_1 = (1/\sqrt{8}) \sum_{m \in \mathcal{O}_1} \phi_m$ and $u_2 = (1/2) \sum_{m \in \mathcal{O}_2} \phi_m$. Their densities add coherently rather than cancelling, so $K_{A_1} = U_{A_1}^T K U_{A_1}$ is a generic 3×3 symmetric matrix. Its off-diagonal norm is 8.9×10^{-2} and its diagonal standard deviation is 2.9×10^{-2} , both nonzero at the level of the interaction. The color algebra is therefore broken to a 2-dimensional abelian subalgebra of the specific form s_0, s_1 above.

V. DISCUSSION

A. Two different gauge-symmetry patterns

The two operators act on the same 14-mode multiplet and yet preserve different subalgebras of $\mathfrak{g}_{\text{SM}}$. The loop preserves a 3-dimensional abelian algebra derived from its class-based eigenvalue partition. The DNLS Hamiltonian preserves a 10-dimensional algebra derived from a coherent cancellation of density overlaps on the B_1 isotypic.

The structural origin of the difference is that the loop's class partition cuts across the D_4 orbit structure (Remark 1), whereas the DNLS Hartree matrix is D_4 -invariant. It is the D_4 -invariance of K combined with the specific real-form structure of the B_1 vectors that forces the exact cancellation in Lemma 7.

B. Independence from calibrated inputs

Neither theorem depends on the DME constant γ , the warp factor kL , or any other calibrated input. The class partition comes from the winding spectrum $\{2\sqrt{2}, 6\sqrt{2}, 0\}$, a pure property of the mode coordinates. The gauge generators come from the D_4 isotypic decomposition. The dimension of each intersection follows from the singular-value gap of the commutator matrix, which is 13 to 15 orders of magnitude and stable across thresholds. The U -independence of the DNLS result was verified by direct computation for $U \in \{0, 0.1, 0.54, 1, 5, 100\}$, with commutator norms at machine precision throughout.

C. Physical interpretation

The loop operator is the WIN processing operator: it extracts particle content from the substrate via a five-layer composition of winding phase, dimensional equilibrium, gradient-flow irreversibility, effective-field-theory damping, and processor projection. Its gauge-symmetry content is minimal: only a rotated 3-dimensional abelian subalgebra survives, and no individual physical charge (hypercharge, weak isospin, color) is preserved.

The DNLS Hamiltonian is the substrate's mean-field time-evolution operator. Its gauge-symmetry content is larger and more structured: the electroweak factor $\mathfrak{su}(2)_L$ is preserved exactly, as are the five isotypic projectors; the color factor $\mathfrak{su}(3)_c$ is broken to a 2-dimensional abelian subalgebra.

The asymmetry is consistent with the WIN Paradigm's interpretation of the two operators: the loop is a readout that reduces the substrate to its gauge-weight content, while the DNLS Hamiltonian is the physical evolution that preserves a specific subalgebra of the gauge structure.

D. Open problems

The following are natural follow-ups:

- Extension of both theorems to the full 56-dimensional Hodge sector $\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$.
- The relation to the A_4 generation structure of the flavor sector [9]: does the A_4 factor commute with M_{loop} and H_{DNLS} when extended to $168 = 3 \times 56$ states?
- The physical interpretation of the specific color subalgebra $\mathfrak{c} = \langle s_0, s_1 \rangle$, and whether it corresponds to a known flavor quantum number.
- The extension of the DNLS Hartree analysis beyond the mean-field approximation, to determine whether the exact interacting Hamiltonian preserves a larger subalgebra.
- A closed-form trigonometric representation of the principal angles, exploiting the algebraic structure of the nullspace basis vectors.

VI. CONCLUSION

We have computed the gauge-symmetry structure of the WIN loop operator M_{loop} and the DNLS Hartree Hamiltonian H_{DNLS} on the 14-mode multiplet at $\lambda = 4$ on $\mathbb{Z}_8 \times \mathbb{Z}_8$. M_{loop} preserves a 3-dimensional abelian subalgebra of $\mathfrak{g}_{\text{SM}}$,

rotated by angles 26.2° , 42.2° , 90.0° from the standard Cartan, and containing no root generator. H_{DNLS} preserves a 10-dimensional subalgebra isomorphic to $\mathfrak{su}(2) \oplus \mathfrak{u}(1)^7$, containing the full $\mathfrak{su}(2)_L$ factor, all five isotypic projectors, and a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$ rotated by 33.4° and 77.8° from the standard color Cartan. The $\mathfrak{su}(2)_L$ preservation is exact and structural, following from a doubled-momentum pairing of modes in the B_1 isotypic that produces exact cancellation of the Hartree density overlaps. Neither result depends on γ , kL , or any calibrated input.

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                 gauge algebra with the commutant of the WIN loop
                 operator  $M_{\text{loop}}$  and the DNLS Hartree Hamiltonian
                  $H_{\text{DNLS}}$  on the 14-mode multiplet at  $\lambda=4$  on
                  $\mathbb{Z}_8 \times \mathbb{Z}_8$ . The loop preserves a rotated
                 3-dimensional abelian subalgebra; the DNLS
                 preserves a 10-dimensional subalgebra isomorphic
                 to  $\mathfrak{su}(2) + \mathfrak{u}(1)^7$ . The  $\mathfrak{su}(2)_L$  preservation is
                 proven exact via doubled-momentum pairing. No
                 result depends on calibrated inputs.}
}

```